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$$r: x=y=z-a$$

$$s: \frac{2x-1}{3} = \frac{y+3}{2} = \frac{z-2}{0}$$

$$s: \frac{x-1/2}{3/2} = \frac{y+3}{-2} = \frac{z-2}{0}$$

$$\vec{dr} = (1, 1, 1)$$

$$Pr = (0, 0, a)$$

$$\vec{ds} = (3/2, -2, 0)$$

$$Ps = (1/2, -3, 2)$$

a ELKAR EBAKITEKO.

$$\vec{PrPs} (\vec{dr} \wedge \vec{ds}) = 0$$

$$|M'| = 0$$

$$\vec{PrPs} = \vec{O}Ps - \vec{O}Pr = (1/2, 3, 2) - (0, 0, a) = (1/2, 3, 2-a)$$

$$\begin{vmatrix} 1 & 3/2 & 1/2 \\ 1 & -2 & 3 \\ 1 & 0 & 2-a \end{vmatrix} = 0$$

$$-2(2-a) + \frac{9}{2} + 1 - \frac{3}{2}(2-a) = 0$$

$$-4 + 2a + \frac{9}{2} + 1 - \frac{6}{2} + \frac{3a}{2} = 0$$

$$\frac{7a-21}{2} = 0 \rightarrow \boxed{a=3}$$

Ebakitze puntua aurkitzeko

parametriketara jotzeko doct $\rightarrow$ **parametro**
desbordinapots

$$r: \begin{cases} x = 0 + \lambda \\ y = 0 + \lambda \\ z = 3 + \lambda \end{cases}$$

$$s: \begin{cases} x = 1/2 + 3/2 \mu \\ y = -3 - 2\mu \\ z = 2 \end{cases}$$

$$x \rightarrow \begin{cases} \lambda = 1/2 + 3/2 \mu \\ \lambda = -3 - 2\mu \\ 3 + \lambda = 2 \end{cases} \Rightarrow$$

$$\begin{cases} 1 - 3/2 \mu = 1/2 \\ \lambda + 2\mu = -3 \\ \lambda = -1 \end{cases}$$

Sistema ebakitz:

$\Rightarrow$ Beraz ebakitze puntua:

$$\boxed{\lambda = -1} \rightarrow \boxed{\mu = -1}$$

$$r: \rightarrow \begin{cases} x = -1 \\ y = -1 \\ z = 2 \end{cases}$$

$$\boxed{(-1, -1, 2)}$$

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$$r: \begin{cases} x = 5 + 4d \\ y = 3 + d \\ z = -d \end{cases}$$

$$s: \frac{x}{m} = \frac{y-1}{3} = \frac{z+3}{n}$$

Kalkulatu m eta n , parallelak izaneko.
Parallelak izoteko $\vec{ds}$ eta $\vec{dr}$ proportionalak
izan behar dira

$$\vec{dr} = (4, 1, -1)$$

$$\vec{ds} = (m, 3, n)$$

$$\frac{4}{m} = \frac{1}{3} = \frac{-1}{n}$$

$$\boxed{12 = m} \quad \boxed{n = -3}$$

Kasu honetan parallelak eta batak epin daturik
Aztertzen epiten da ea $Pr \notin S$.

$$Pr(5, 3, 0) \rightarrow s: \frac{5}{12} = \frac{3-1}{3} = \frac{0+3}{-3}$$

$$\frac{5}{12} \neq \frac{2}{3} \neq -1$$

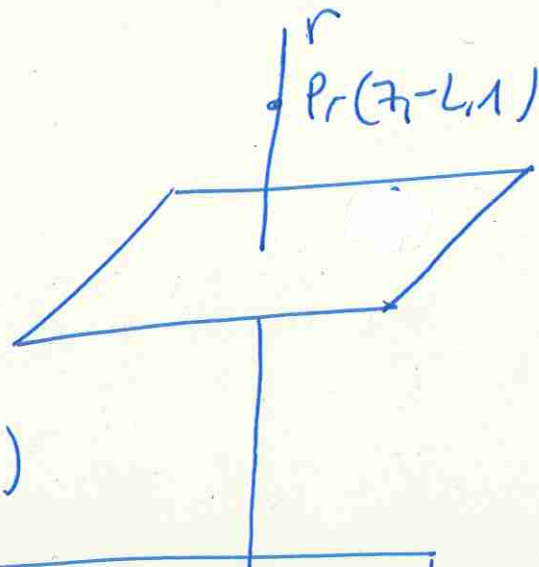
Beraz $Pr \notin S \rightarrow \boxed{\text{PARALELOAK dira}}$

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$$\pi: x+2y=5$$

$$r \perp \pi$$

$$P_r(7, -2, 1) \in r$$



$$r \text{ zuzenoreu } \bar{dr} = \bar{n}_\pi = (1, 2, 0)$$

$$r: \begin{cases} \bar{dr}(1, 2, 0) \\ P_r(7, -2, 1) \end{cases}$$

 $\Rightarrow$

$$r: \begin{cases} x = 7 + d \\ y = -2 + 2d \\ z = 1 \end{cases}$$

ez dago

$$P_s(-1.5, 0)$$

$$S \parallel r: \begin{cases} x+y=1 \\ y-z=2 \end{cases}$$

$$S: \begin{cases} \bar{ds} = \bar{dr} = \bar{n}_1 \wedge \bar{n}_2 = (1, 1, 0) \wedge (0, 1, -1) \\ P_s(-1.5, 0) \end{cases}$$

$$\bar{dr} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 0 \\ 0 & 1 & -1 \end{vmatrix} = -\hat{i} + \hat{k} + \hat{j} = (-1, 1, 1)$$

$$S: \begin{cases} x = -1 + d \\ y = 5 + d \\ z = 0 + d \end{cases}$$